2. Project Proposal

Executive summary

Electric Vehicle Market Performance & Insights is a data-visualization project based on the supplied `ev_market_test.csv`. The file contains 100,000 records and 37 columns covering vehicle identity, manufacturing geography, market region, category, battery and powertrain specifications, charging, price and insurance, warranty, safety/autonomy indicators, customer rating, annual sales units, and a CO■-savings field. The project uses descriptive analysis and Tableau to make these attributes easier to compare.

Rationale

The dataset is wide and contains both numeric and categorical variables. A curated dashboard can provide an overview while a story can guide the audience through selected comparisons. The website gives the analysis a single presentation point.

Deliverables

1. Dataset profile and field dictionary.
2. Data-quality audit and cleaning log.
3. Exploratory analysis and KPI definitions.
4. Tableau worksheets and dashboard.
5. Tableau story with evidence-backed narrative.
6. Responsive `index.html` integrating the views.
7. Test log, user guide, final report, and demonstration script.

Dataset summary

- Records: 100,000; fields: 37.
- Model years: 2018–2026.
- 200 manufacturers, 10 manufacturing countries, 7 market regions.
- 15 vehicle categories and 5 battery types.
- Missing cells: 9,511; duplicate complete rows: 0; duplicate vehicle IDs: 0.

Method

Profile → validate → document missingness and outliers → prepare analysis-ready fields → perform exploratory analysis → build and validate Tableau views → publish → embed and test.

Constraints and assumptions

No source citation or data-generation methodology was included with the CSV. Therefore, the report must describe this as analysis of the supplied dataset and avoid asserting that it represents all EVs, real sales, or actual measured emissions unless provenance is later verified.

Risks

- Ambiguous definitions for sales, CO■ saved, cycle life, charging times, and customer rating.
- Missing values in four fields.
- Extreme values in price, sales, efficiency, and battery capacity.
- Tableau embed behavior depending on public permissions and internet access.